

Conversational Product Recommendation System using LLM

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Abstract- — The Conversational Product Recommendation System using LLM is an AI-driven application designed to enhance product recommendation by enabling natural language interaction between users and the system. In modern e-commerce environments, users often face information overload due to the vast number of available products. Traditional recommendation systems rely on static filtering methods and fail to understand complex user queries expressed in natural language. To address these limitations, the proposed system integrates Large Language Models (LLMs) with Natural Language Processing (NLP) techniques to interpret user intent, preferences, and constraints. The system provides a chatbot-based interface where users can interact conversationally, refine their queries, and receive personalized product recommendations in real time. A recommendation engine processes extracted features and ranks products based on relevance, while a backend database manages product data and user interactions.

Keywords:- Conversational AI, Product Recommendation System, Large Language Models (LLM), Natural Language Processing (NLP), Personalization, Chatbot, E-commerce, Recommendation Engine.

I. INTRODUCTION

In today's digital era, e-commerce platforms have become an essential part of everyday life, offering users a wide variety of products across multiple categories. While this abundance of choices provides flexibility, it also creates a challenge known as information overload, where users find it difficult to identify the most suitable product based on their needs and preferences. Traditional product recommendation systems, such as collaborative filtering and content-based filtering, are widely used to assist users in product selection. However, these systems have limitations as they rely heavily on historical data, predefined filters, and user ratings. They often fail to understand complex user requirements expressed in natural language and lack the ability to interact dynamically with users. The project titled "Conversational Product Recommendation System using LLM" aims to overcome the limitations of traditional systems by integrating conversational AI with recommendation techniques.

This study focuses on integrating Natural Language Processing (NLP) into recommendation systems to enable conversational interaction. The authors highlight that traditional systems lack the ability to understand user queries expressed in natural language.

2. Product Recommendation Using Deep Learning Models Authors: Chen, Y., & Zhao, L. (2023)

This paper presents the use of deep learning techniques for enhancing product recommendation systems. The authors propose models that analyze user behavior, preferences, and product features to generate personalized suggestions.

3. Large Language Models for Intelligent Chatbots Authors: OpenAI Research Group (2023)

This research explores the capabilities of Large Language Models (LLMs) in building intelligent chatbot systems. It explains how LLMs can understand context, process natural language, and generate human-like responses.

4. Hybrid Recommendation Systems for E-commerce Authors: Kumar, A., & Singh, R. (2022)

This paper discusses hybrid recommendation systems that combine collaborative filtering and content-based filtering techniques. The authors highlight that individual methods have limitations, such as cold-start problems and lack of diversity.

II. LITERATURE SURVEY

Early Works

1. Conversational Recommender Systems Using Natural Language Processing Authors: Brown, T., & Williams, S. (2022)

Objectives

The primary objectives of this project include:

- To develop a conversational product recommendation system using LLM.
- To enable users to interact using natural language.
- To understand user intent and preferences accurately.
- To generate personalized product recommendations.
- To provide a real-time interactive chatbot interface.

III. METHODOLOGY

The system integrates real-time data collection, processing, and visualization techniques to analyze website traffic efficiently.

System Workflow

1. Data Collection

Product data is collected from datasets, APIs, or e-commerce platforms, including details like price, features, ratings, and category.

2. Data Preprocessing

The collected data is cleaned and structured by removing duplicates, handling missing values, and normalizing attributes for efficient processing.

3. User Query Input

Users interact with the system through a chatbot interface by entering queries in natural language.

4. NLP Processing

The system processes the input using NLP techniques such as tokenization and text cleaning to prepare it for analysis.

5. LLM-Based Understanding

The Large Language Model analyzes the query to extract user intent, preferences, budget, and required features.

6. Recommendation Generation

The recommendation engine filters and ranks products based on extracted information and retrieves the most relevant results.

7. Response Generation

The system generates a conversational response including product suggestions with details like price, features, and ratings.

8. Continuous Interaction

Users can refine their queries, and the system updates recommendations dynamically in real time.

IV. PROPOSED SYSTEM

The Conversational Product Recommendation System using LLM provides personalized product suggestions through a chatbot interface using natural language. It uses NLP and LLM to understand user queries, extract preferences, and generate relevant recommendations. The system retrieves and ranks products from a database and displays them in real time.

System includes:

- Conversational chatbot interface
- LLM-based query understanding
- Recommendation engine
- Real-time response system

1. Data Collection Phase

Product data is collected from datasets/APIs → Stored in database → Prepared for processing.

2. Processing Phase

- User enters query through chatbot.
- NLP and LLM process the input.
- Extracts user intent, budget, and preferences.

3. Recommendation Phase

- System retrieves relevant products from database.
- Applies filtering and ranking techniques.
- Generates personalized recommendations.

4. Response Phase

- Recommendations are converted into conversational responses.
- Displayed to the user with product details.

Technologies:

Frontend (HTML, CSS, JS), Backend (Flask/Django), Database (MySQL/MongoDB), LLM APIs.

V. APPLICATIONS

The system has wide applications in website management and cybersecurity:

- E-commerce Platforms – Provides personalized product recommendations to users.
- Online Shopping Assistants – Acts as a virtual assistant to help users find products easily.

- Customer Support Systems – Assists customers by suggesting suitable products based on queries.
- Digital Marketing – Helps businesses promote relevant products to targeted users.
- Mobile Shopping Apps – Enhances user experience with real-time conversational recommendations.
- Personalized Recommendation Services – Can be used in domains like electronics, fashion, and books.

VI. ALGORITHMS

The system uses multiple algorithms for efficient traffic monitoring and analysis.

1. Data Collection & Preprocessing Algorithm

Purpose: Collect and prepare product data for recommendation.

Steps:

Collect product data from datasets/APIs.

Remove duplicates and handle missing values.

Normalize product attributes (price, ratings, features). Store processed data in database.

2. User Query Processing Algorithm

Purpose: Understand user input using NLP and LLM. Steps:

Accept user query through chatbot.

Perform text preprocessing (tokenization, cleaning). Analyze query using LLM.

Extract user intent, preferences, and constraints.

3. Recommendation Generation Algorithm Purpose: Generate relevant product suggestions. Steps:

Retrieve product data from database.

Match user preferences with product features. Apply filtering techniques.

Rank products based on relevance.

4. Response Generation Algorithm

Purpose: Present recommendations in conversational format.

Steps:

Select top-ranked products.

Add product details (price, features, ratings).

Generate natural language response using LLM. Display results to user via chatbot.

5. Continuous Interaction Algorithm

Purpose: Improve recommendations through user interaction. Steps:

Accept follow-up queries from user.

Update preferences based on new input. Regenerate recommendations.

Repeat process for real-time interaction.

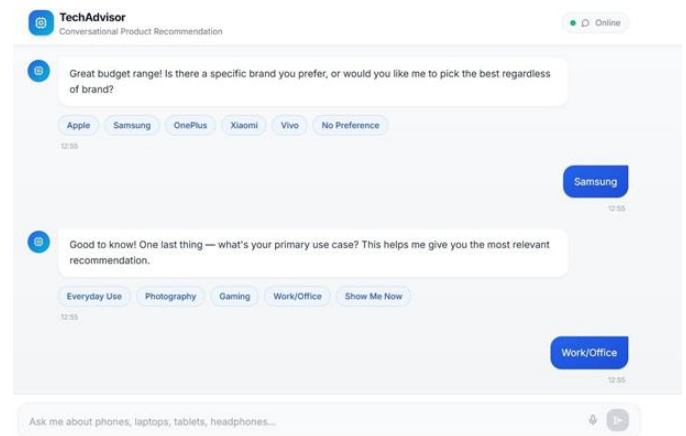


Fig 1: Taking Input

VII. RESULT

The Conversational Product Recommendation System using LLM was successfully developed and implemented, demonstrating its effectiveness in delivering intelligent and personalized product recommendations. The system is capable of understanding user queries expressed in natural language and accurately extracting key information such as product category, budget, preferences, and required features using NLP and LLM techniques.

The chatbot-based interface provides a smooth and interactive user experience, allowing users to communicate with the system in a natural and intuitive manner. Users can refine their queries through continuous interaction, and the system dynamically updates recommendations in real time, making the process highly flexible and user-friendly.

Output Screen 1:-

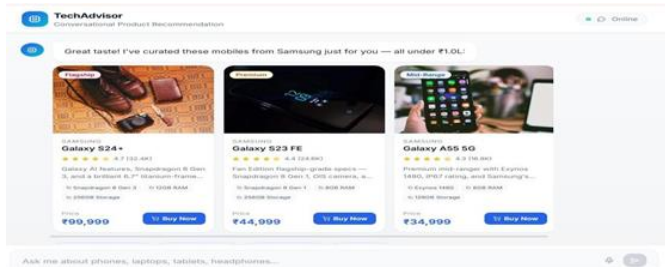


Fig 2: Output Screen 1(Home page)

Output Screen 2:-

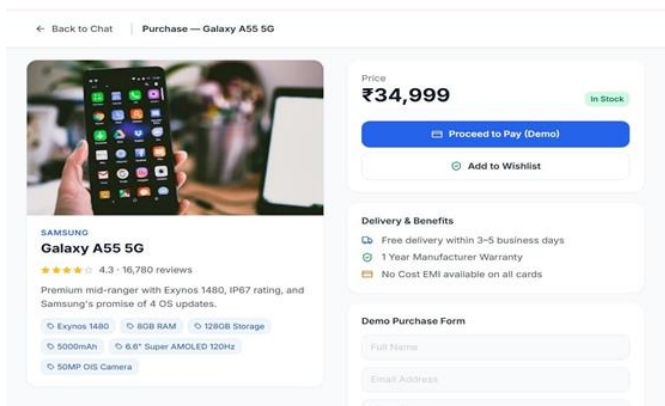


Fig 3: Output Screen 1(Dashboard page)

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VIII. CONCLUSION

The Smart and Intelligent Web Traffic Analytics and Monitoring System successfully provides a robust platform for tracking and analyzing website visitor data in real-time. By integrating a dynamic dashboard with automated tracking scripts, the system enables administrators to gain clear insights into user behavior and identify traffic patterns effectively. The inclusion of intelligent anomaly detection adds a critical layer of security, allowing for the immediate identification of suspicious activities that could threaten website integrity. Ultimately, this project simplifies complex data through intuitive visualizations and charts, empowering users to make informed decisions that enhance both site performance and overall security. It serves as a comprehensive tool that bridges the gap between raw web logs and actionable business intelligence.